

Lab 1: Basic Logic Design

Requirements Not Met

N/A

Problems Encountered

Getting Logisim to run was difficult; it would not open until I restarted my computer. I also struggled to install Quartus and Questa on Arch Linux.

Applications

The content of this lab is extremely relevant to describing decision-making algorithms with digital signals. Any digital system that combines multiple inputs uses logical gates like these to do so; they are the basis of digital computing. The FPGA portion is also very applicable, as FPGAs are used widely in DSP, to manipulate data at high rates via the programmable circuitry.

Pre-Lab Questions

Part 1: Boolean algebra simplification

a. $F = A * B + A * \overline{B} + \overline{A} * B$

$$F = (A * B) + (A * \overline{B}) + (\overline{A} * B)$$

$$F = A * (\overline{B} + B) + (\overline{A} * B)$$

$$F = A * 1 + (\overline{A} * B)$$

$$F = A + (\overline{A} * B)$$

$$F = (A + \overline{A}) * (A + B)$$

$$F = 1 * (A + B)$$

$$F = A + B$$

Order of operations

Inverse distribution

Inverse

Identity

Distribution

Inverse

Identity

b. $Z = \overline{V}W + X + \overline{Y}(X + \overline{V}W)$

$$Z = X + \overline{V}W + (X + \overline{V}W)\overline{Y}$$

$$Z = (X + \overline{V}W) + (X + \overline{V}W)\overline{Y}$$

$$Z = X + \overline{V}W$$

Commutation

Association

Absorption

Part 2: Implementation

	Unsimplified	Simplified
F		
Z		

Part 3: Security system design

a.

D	W	M	A	S
0	0	0	0	1
0	0	1	1	1
0	1	0	0	1
0	1	1	1	1
1	0	0	0	1
1	0	1	1	1
1	1	0	0	0
1	1	1	0	1

b. $A(D, W, M) = \Sigma_m(1, 3, 5) = \overline{D}\overline{W}M + \overline{D}WM + D\overline{W}M$

Simplify:

$$\begin{aligned}
 &\overline{D}(\overline{W}M + WM) + D\overline{W}M \\
 &\overline{D}(M\overline{W} + MW) + D\overline{W}M \\
 &\overline{D}(M(\overline{W} + W)) + D\overline{W}M \\
 &\overline{D}(M \cdot 1) + D\overline{W}M \\
 &\overline{D}M + D\overline{W}M \\
 &M(\overline{D} + D\overline{W}) \\
 &M((\overline{D} + D)(\overline{D} + \overline{W})) \\
 &M(1 \cdot (\overline{D} + \overline{W})) \\
 &M(\overline{D} + \overline{W}) \\
 &\overline{MDW}
 \end{aligned}$$

Inverse distribution
Commutation
Inverse distribution
Inverse
Identity
Inverse distribution
Distribution
Inverse
Identity
De Morgan's

So, $A = M\overline{D} \cdot \overline{W}$

c. $S(D, W, M) = \Pi_m(6) = \overline{D} + \overline{W} + M$

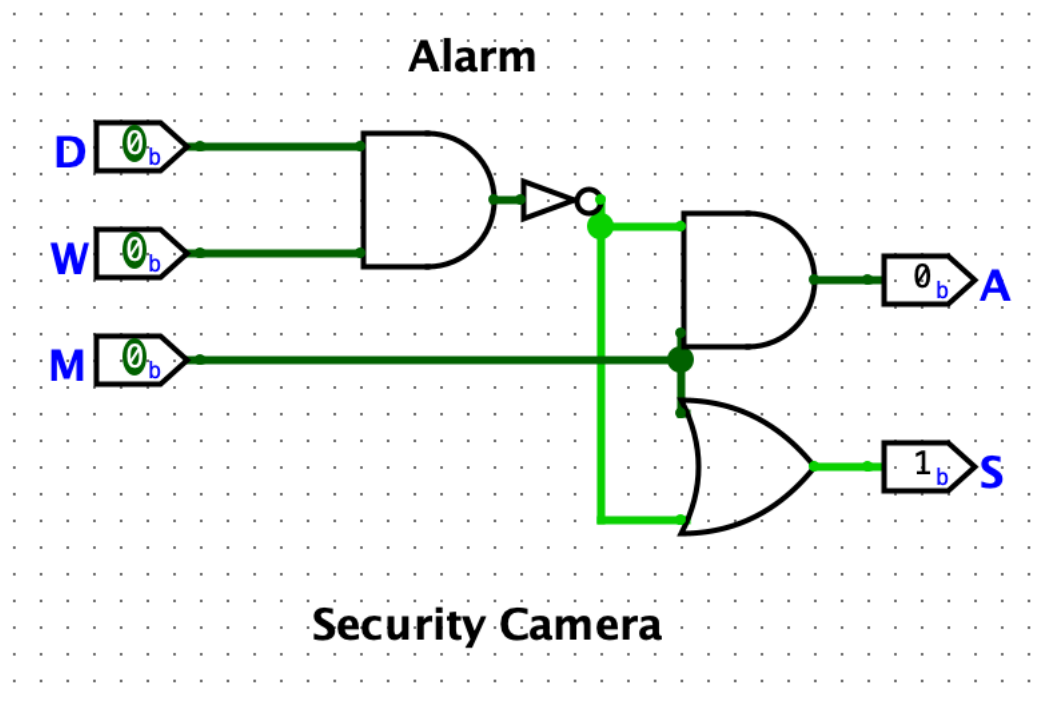
Simplify:

$$\overline{D\overline{W}} + M$$

De Morgan's

So, $S = \overline{D \cdot W} + M$

d. Implementation:



Note that the two circuits are combined.